

Milestone 3: User Feedback Reflection

1. Questions Asked and Aspects Evaluated

To evaluate the usability and effectiveness of the Vancouver Crime Dashboard, a survey was distributed to users who interacted with the dashboard. The questions were designed to assess several aspects of the application, including usability, visual design, performance, and usefulness of the information.

The survey included questions about:

User background

- Where the user is from (urban, suburban, rural, or small town)
- Whether they have used similar crime or data dashboards before

Usability and navigation

- Ease of navigating different sections of the dashboard
- How intuitive the interactive features are (filters, tooltips, and map interactions)
- Which dashboard features users used the most

Visual design

- How visually appealing the charts and maps are
- Whether the colors, text, and layout are readable and accessible

Information clarity

- How easy it is to understand the information presented in the visualizations

Usefulness

- Whether the dashboard helps users understand crime patterns in Vancouver

- Whether it could help users make decisions about where to live or visit

Performance

- Satisfaction with the dashboard's loading speed and responsiveness

Open-ended questions

- What users liked most about the dashboard
- What users liked least
- What features could be improved
- Any additional comments or suggestions

These questions allowed both quantitative ratings and qualitative feedback to better understand how users experience the dashboard.

2. Summary of Feedback Received

Four users completed the survey.

User background

All respondents reported that they are from urban areas, and none had previously used similar crime or data dashboards. This suggests that the dashboard was evaluated primarily by users without prior experience with this type of visualization tool.

Navigation and usability

Navigation feedback was very positive.

- 75% of users rated navigation as "Very easy"
- 25% rated it as "Neutral"

This suggests that the structure and organization of the dashboard make it easy for most users to move between sections and features.

Interactive elements also received positive ratings:

- 50% rated the interactive elements as “Very easy to use”
- 25% rated them as “Easy to use”
- 25% rated them as “Neutral”

Overall, users found the filters, tooltips, and map interactions generally intuitive.

Visual design

The visual appeal of the charts and maps was rated positively:

- 66.7% rated them as “Very appealing”
- 33.3% rated them as “Appealing”

However, several users noted that many charts share the same color and style, which may reduce visual distinction between different charts.

Understanding the visualizations

Users generally found the visualizations easy to interpret.

- 75% rated them as “Easy to understand”
- 25% rated them as “Very easy to understand”

This indicates that the visualizations successfully communicate the underlying data and trends.

Usefulness of the dashboard

Users felt that the dashboard provides useful insights into Vancouver crime patterns.

- 75% rated the dashboard as “Very useful”
- 25% rated it as “Neutral”

When asked whether the dashboard could help them make decisions about where to live or visit in Vancouver:

- 50% responded “Yes”
- 50% responded “Definitely”

This suggests that the dashboard provides meaningful insights that could support real-world decision-making.

Performance and responsiveness

Performance feedback was mixed.

- 25% rated the dashboard as “Very fast”
- 25% rated it as “Fast”
- 25% rated it as “Acceptable”
- 25% rated it as “Slow”

Additionally, one user specifically mentioned that zooming in and out on the map can feel slow, indicating that the interactive map may require performance optimization.

Feature usage

The most frequently used feature was the map view, which 75% of users reported using the most.

The summary panel was used most by the remaining 25% of users.

This indicates that the interactive map is the primary feature that users engage with when exploring the dashboard.

Open-ended feedback

Users highlighted several strengths and areas for improvement.

What users liked most

The most commonly mentioned positive feature was the interactive map, with users specifically mentioning:

- The ability to switch between overall map view and section map view
- The clarity of the map visualization
- The interactivity of the map

This reinforces that the map is the most valuable component of the dashboard.

What users liked least

Several users mentioned that:

- The bar charts use the same colors and styles
- The bar chart visualizations could be improved

This suggests that chart design could be enhanced to improve visual variety and clarity.

Suggested improvements

Users suggested several potential improvements:

- Add color indicators to highlight higher values in bar charts
- Improve the loading speed of the interactive map
- Adjust the dashboard layout, particularly making the right column wider
- Remove or reduce unused sidebar space

One user also commented that the dashboard is generally good overall, indicating that the issues identified are relatively minor.

3. Planned Improvements Based on Feedback

Based on the user feedback collected, several improvements will be implemented to enhance the dashboard's usability and performance.

Improve chart visual design

Because users mentioned that many charts use similar colors and styles, future revisions will:

- Introduce more varied color palettes across charts
- Use color highlighting to emphasize higher values in bar charts
- Adjust styling to improve visual differentiation between charts

These changes will help users interpret charts more easily and improve visual engagement.

Improve map performance

Since some users reported slow map interactions, particularly during zooming, the following improvements will be explored:

- Optimizing the amount of data rendered on the map
- Using clustering or aggregation techniques
- Improving rendering efficiency for map layers

Improving performance will make the map more responsive and enhance the overall user experience.

Adjust dashboard layout

One user suggested that the right column could be wider and that some sidebar elements may be unnecessary. Future revisions may:

- Rebalance the dashboard layout to better use screen space
- Increase the width of chart areas
- Simplify or reduce unused interface elements

These adjustments should improve readability and visual balance.

Maintain the dashboard's strengths

While implementing improvements, several successful aspects of the dashboard will be preserved:

- The interactive map, which users found most useful
- The clear structure and navigation of the dashboard
- The ability to switch between overall and sectional map views

These features represent the core value of the dashboard and will remain central to the design.